Day:1

notes

what is a variable in python?

In Python, a variable is a name you give to a piece of data so you can store it and use it later in your program.

Think of a variable like a labeled box:

The label is the variable name The content inside the box is the value

Example:

age = 15

name = "Alex"

age is a variable storing the number 15

name is a variable storing the text "Alex"

example:

```
print("10").l+
```

```
10
```

Clean code techniques

1:-Readabality

2:-Reusability

3:-Time complexity

4:-

Time Complexity

Time complexity tells us how fast or slow a program runs as the input size increases.

It measures the number of operations an algorithm performs.

We usually write it using Big O notation.

Example 1

— O(n)

arr = [1, 2, 3, 4, 5]

for x in arr:

```
print(x)
```

If the array has:

5 elements → loop runs 5 times

100 elements → loop runs 100 times

So the time grows linearly → O(n)

Example 2

— $O(n^2)$

for i in range(n):

```
for j in range(n):
```

```
    print(i, j)
```

Here:

Outer loop runs n times

Inner loop also runs n times

Total operations = n × n = n^2

So complexity is:

O(n2)

What is TLE (Time Limit Exceeded)?

"TLE means your program took too much time to run and exceeded the allowed time limit."

This usually happens in:

:-Coding contests

:-LeetCode

:-HackerRank

:-CodeChef

:-Interviews

Example:-

Suppose the time limit is:

1 second

If your algorithm performs:

1,000 operations → OK ✅

10,000,000,000 operations → Too slow ❌

Then the platform shows:

TLE — Time Limit Exceeded

Why TLE Happens?

Common reasons:

Using nested loops unnecessarily

Using slow algorithms

Repeating calculations

Large input size with inefficient code

if conditions

In Python, if conditions are used to make decisions in a program.

They allow the program to execute certain code only when a condition is true.

Basic Syntax

if condition:

```
# code to execute
```

Example 1

age = 18

if age >= 18:

```
print("You are eligible to vote")
```

Output:

You are eligible to vote

Since the condition age >= 18 is true, the message is printed.

```
Start coding or generate with AI.
```

what is type casting in python?

In Python, type casting means converting one data type into another data type.

For example:

Convert a string → integer

Convert integer → float

Convert float → string

Why Type Casting is Needed

Sometimes data comes in the wrong format.

Example:

age = "18"

Here "18" is a string, not a number.

If you want mathematical operations, you must convert it to an integer.

Example of Type Casting

age = "18"

new_age = int(age)

print(new_age + 2)

Output

20

what is type conversion in python?

In Python, type conversion means changing one data type into another data type automatically or manually.

Example:

Integer → Float

String → Integer

Float → String

Types of Type Conversion

There are 2 types:

Implicit Type Conversion (automatic)

Explicit Type Conversion / Type Casting (manual)

1. Implicit Type Conversion

Python automatically converts one type into another when needed.

Example:-

x = 10

y = 2.5

result = x + y

print(result)

print(type(result))

Output

12.5

<class 'float'>

What happened?

x is an integer

y is a float

Python automatically converts 10 into 10.0

So the result becomes a float.

2. Explicit Type Conversion (Type Casting)

The programmer manually converts the type using functions.

Example:-

num = "100"

converted_num = int(num)

print(converted_num)

print(type(converted_num))

Output:-

100

<class 'int'>

Main Difference

Type Conversion | | Type Casting

General process | | Manual process

Can be automatic or manual | | Always manual

Done by Python or programmer | | Done only by programmer

Includes implicit conversion | | Uses functions like int(), float()

array

```
n=int(input())
for i in range(n):
    a,b,c=list(map(int,input().split()))
    print(a,b,c)
```

```
2
1 2 3
1 2 3
1 2 3
1 2 3
```

```
a=3
b=4.3
c=a*b
print(c)
print(int(c))
```

```
12.899999999999999
12
```

```
T=int(input())
for i in range(T):
    n=int(input())
    k=int(input())
    print(k%n)
```

```
2
3
14
2
10
100
0
```

```
a, b = map(int, input().split())
if a < b:
    print("a < b")
elif a > b:
```

```
    print("a > b")
else:
    print("a == b")
```

```
4 4
a == b
```

```
# name=input()
# name,place=input().split()
# a,b=map(input().split)
# a,b= map(int,input()).split()
# a,b= map(int,(input()).split())
a,b= map(float,(input()).split())
print(a,b)
```

```
2 3
2 3
```

```
for i in range(0,9):
  for j in range(i):
    print(j+1, end="")
  print()
```

```

1
12
123
1234
12345
123456
1234567
12345678
```

```
size = 5
for i in range(size):
    for j in range(10):
        if i == j or i == size -  1 or j == 0 or j == 10  -1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

```
*        *
**       *
* *      *
*  *     *
**********
```

```
student_data=["akhil","pulivendula","akhilboggiti@gmail.com",6303985957]
student_data[1]="akhil"
print(student_data)
```

```
['akhil', 'akhil', 'akhilboggiti@gmail.com', 6303985957]
```

```
names=["h","e","l","l","o"]
# names.append(1)  #adds at end
# names.insert(1,2) #adds element at specific index
# names.extend(["w","o","r","l","d"])
# names.pop() removes element using index
# names.remove("l")
# names.clear()
# names.reverse()
# names.copy
# names.count

names.index("l")
print(names)
```

```
['h', 'e', 'l', 'l', 'o']
```

```
Akhil=[1,2,3,4]
for akhil in Akhil:
  if(akhil>1 and akhil<4):
    print(akhil)
```

```
2
3
```

Double-click (or enter) to edit

Start coding or generate with AI.